

HIV TRIALS

1. GEMINI 1 & 2 Trials

- **Objectives:** To demonstrate that a two-drug regimen of dolutegravir (DTG) + lamivudine (3TC) was as effective (non-inferior) as a standard three-drug regimen of DTG + tenofovir disoproxil fumarate/emtricitabine (TDF/FTC) for treatment-naïve adults with HIV-1.
- **Findings:** The two-drug regimen was non-inferior to the three-drug regimen in achieving virologic suppression (HIV-1 RNA <50 copies/mL) at 48 weeks. High efficacy was maintained through week 144 in both treatment groups.
- **Recommendations:** These trials provided evidence that a two-drug regimen can be a viable first-line treatment, potentially reducing the lifetime drug exposure for patients with HIV.
- **Medications:** Dolutegravir (DTG), Lamivudine (3TC), Tenofovir disoproxil fumarate (TDF), Emtricitabine (FTC).
- **Publication:** *Journal of Acquired Immune Deficiency Syndromes (JAIDS) & Open Forum Infectious Diseases*.
- **Year:** 2019 (48-week results); 2025 (144-week analysis).
- **Participants:** 1,441 ART-naïve adults with HIV-1 infection (719 in GEMINI-1 and 722 in GEMINI-2).

2. START Trial (Strategic Timing of AntiRetroviral Treatment)

- **Objectives:** To determine whether starting antiretroviral therapy (ART) immediately in asymptomatic, HIV-positive adults with a CD4+ count >500 cells/mm³ is superior to deferring treatment until the CD4+ count drops to <350 cells/mm³.
- **Findings:** Immediate ART significantly reduced the risk of the primary composite endpoint of serious AIDS events, serious non-AIDS events, or death by 57% compared to deferred treatment.
- **Recommendations:** The findings provided the pivotal evidence for a global consensus to recommend that all people living with HIV should start ART immediately upon diagnosis, regardless of their CD4+ count.
- **Medications:** Various standard-of-care ART regimens were used, determined by local guidelines and physician choice.
- **Publication:** *New England Journal of Medicine (NEJM)* and *NEJM Evidence*.
- **Year:** 2015 (primary report); 2023 (long-term follow-up).
- **Participants:** 4,684 ART-naïve, HIV-positive adults with a CD4+ count >500 cells/mm³.

3. TANGO Trial

- **Objectives:** To assess the non-inferior antiviral activity of switching to a two-drug regimen of DTG + 3TC compared to continuing a standard tenofovir alafenamide (TAF)-based regimen of 3 or more drugs in virologically suppressed adults with HIV-1.
- **Findings:** Switching to DTG/3TC was found to be non-inferior to continuing the TAF-based regimen in maintaining virologic suppression (HIV-1 RNA <50 copies/mL) through 48 weeks.
- **Recommendations:** This study supported the safety and efficacy of switching to a simpler, two-drug maintenance regimen for individuals who are already virologically suppressed.
- **Medications:** Dolutegravir (DTG), Lamivudine (3TC), Tenofovir Alafenamide (TAF)-based regimens.
- **Publication:** *Open Forum Infectious Diseases*.
- **Year:** 2024.
- **Participants:** Virologically suppressed, ART-experienced adults. Specific enrollment numbers are not detailed in the provided text, but it is a Phase 3 trial.

4. HPTN 052 Trial

- **Objectives:** To evaluate if antiretroviral therapy (ART) can prevent the sexual transmission of HIV among serodiscordant couples, and to determine the optimal time to start ART.
- **Findings:** Early initiation of ART in the HIV-infected partner led to a 96% reduction in genetically-linked HIV transmission to their uninfected partner.
- **Recommendations:** This landmark study provided proof-of-concept for "Treatment as Prevention" (TasP), leading to the WHO recommendation in 2013 that ART be offered to all HIV-positive individuals in a serodiscordant relationship.
- **Medications:** A combination of 11 HIV drugs from various classes, including Atazanavir, Efavirenz, Emtricitabine/TDF, Lamivudine, Lopinavir/ritonavir, and Nevirapine, were available as study drugs.
- **Publication:** *New England Journal of Medicine* (NEJM).
- **Year:** 2011 (primary results); 2016 (final results).
- **Participants:** 1,763 HIV serodiscordant couples, with the majority (97%) being heterosexual. The HIV-infected partners had a CD4+ count between 350-550 cells/mm³ at enrollment.

5. iPrEx Trial (Pre-Exposure Prophylaxis Initiative)

- **Objectives:** To evaluate the safety and efficacy of a once-daily oral pill containing emtricitabine (FTC) and tenofovir disoproxil fumarate (TDF) for HIV pre-exposure prophylaxis (PrEP) in men who have sex with men (MSM) and transgender women.
- **Findings:** The daily FTC/TDF pill reduced the risk of HIV acquisition by 42% compared to placebo. Efficacy was strongly correlated with drug adherence, rising to over 90% in those with detectable drug levels.
- **Recommendations:** The trial established the first proof that oral PrEP could be effective, leading to the approval of FTC/TDF (Truvada) for HIV prevention in the United States and a paradigm shift in global HIV prevention strategies.
- **Medications:** Emtricitabine (FTC) and Tenofovir disoproxil fumarate (TDF) as a single combination pill.
- **Publication:** *New England Journal of Medicine* (NEJM).
- **Year:** 2010 (primary results).
- **Participants:** 2,499 HIV-seronegative MSM and transgender women who reported high-risk behavior for HIV-1 infection.

6. Partners PrEP Study

- **Objectives:** To determine the safety and efficacy of daily oral tenofovir (TDF) or daily oral emtricitabine/tenofovir (FTC/TDF) for HIV PrEP in HIV-uninfected partners within heterosexual serodiscordant couples.
- **Findings:** Compared to placebo, the risk of HIV acquisition was reduced by 62% in the TDF group and by 73% in the FTC/TDF group. Both drugs were effective for men and women.
- **Recommendations:** This trial extended the evidence for oral PrEP's efficacy to heterosexual populations and provided evidence for the use of a single drug (TDF) as an alternative.
- **Medications:** Tenofovir disoproxil fumarate (TDF), Emtricitabine/Tenofovir disoproxil fumarate (FTC/TDF, brand name Truvada).
- **Publication:** *New England Journal of Medicine* (NEJM).
- **Year:** 2011 (interim results led to placebo discontinuation).
- **Participants:** 4,758 HIV-1 serodiscordant heterosexual couples in Kenya and Uganda.

7. PROMISE Trial (Promoting Maternal and Infant Survival Everywhere)

- **Objectives:** The breastfeeding component aimed to compare the safety and efficacy of maternal triple-drug ART versus daily infant nevirapine (NVP) for the prevention of mother-to-child transmission (PMTCT) of HIV during breastfeeding.
- **Findings:** Both strategies were found to be safe and highly effective, with maternal ART virtually eliminating HIV transmission through breast milk. Overall infant survival at 12 months was extremely high (98.9%) in both arms.
- **Recommendations:** The study provided robust evidence to support WHO guidelines recommending that all pregnant and breastfeeding women living with HIV be offered lifelong ART.
- **Medications:** Maternal triple-drug ART (regimens not specified in the text), Infant Nevirapine (NVP).
- **Publication:** *New England Journal of Medicine* (NEJM) and conference presentations.
- **Year:** 2016 (results announced).
- **Participants:** 2,431 HIV-infected mothers and their HIV-uninfected infants from sub-Saharan Africa and India.

8. SINGLE Trial

- **Objectives:** To evaluate the safety and efficacy of a regimen of dolutegravir (DTG) plus abacavir/lamivudine (ABC/3TC) versus a standard regimen of efavirenz/tenofovir/emtricitabine (EFV/TDF/FTC) in ART-naïve adults.
- **Findings:** The DTG-containing regimen was superior to the EFV-based regimen at week 144, primarily due to fewer participants discontinuing treatment because of adverse events (4% vs. 14%). Virologic suppression rates were 71% for DTG vs. 63% for EFV.
- **Recommendations:** The trial established dolutegravir as a better-tolerated and more durable first-line option compared to an older efavirenz-based standard.
- **Medications:** Dolutegravir (DTG), Abacavir/Lamivudine (ABC/3TC), Efavirenz/Tenofovir/Emtricitabine (EFV/TDF/FTC).
- **Publication:** *Journal of Acquired Immune Deficiency Syndromes* (JAIDS).
- **Year:** 2015 (Week 144 results).
- **Participants:** 833 ART-naïve, HIV-1 positive adults.

9. FLAMINGO Trial

- **Objectives:** To compare the safety and efficacy of a dolutegravir (DTG)-based regimen versus a ritonavir-boosted darunavir (DRV/r)-based regimen, each combined with a background of two NRTIs, in treatment-naïve adults.
- **Findings:** The DTG-based regimen was superior, with 80% of participants achieving an undetectable viral load (<50 copies/mL) at week 96, compared to 68% in the DRV/r group. The difference was most pronounced in patients with a high baseline viral load. Tolerability was similar between groups.
- **Recommendations:** This trial demonstrated that an integrase inhibitor-based regimen (DTG) was more effective than a protease inhibitor-based regimen (DRV/r) for first-line therapy.
- **Medications:** Dolutegravir (DTG), Darunavir (DRV), Ritonavir (r), and investigator-selected NRTIs (tenofovir/emtricitabine or abacavir/lamivudine).
- **Publication:** *The Lancet HIV*.
- **Year:** 2015 (96-week results).
- **Participants:** 484 ART-naïve, HIV-1 infected adults.

10. ADVANCE Trial

- **Objectives:** To compare the efficacy and safety of three first-line antiretroviral regimens in treatment-naïve individuals: TAF/FTC + DTG, TDF/FTC + DTG, and a standard-of-care TDF/FTC/EFV.
- **Findings:** All three regimens showed high viral suppression rates at 192 weeks, with no DTG resistance developing. However, significant weight gain was observed, which was highest in the TAF/FTC + DTG arm (mean 8.9 kg), intermediate in the TDF/FTC + DTG arm (mean 5.9 kg), and lowest in the EFV-based arm (mean 3.2 kg).
- **Recommendations:** This trial supported the high potency of DTG-based regimens but raised awareness of the clinically significant weight gain associated with them, particularly in combination with TAF and among women.
- **Medications:** Dolutegravir (DTG), Tenofovir alafenamide (TAF), Emtricitabine (FTC), Tenofovir disoproxil fumarate (TDF), Efavirenz (EFV).
- **Publication:** *Open Forum Infectious Diseases*.
- **Year:** 2024 (192-week final results).
- **Participants:** 1,053 ART-naïve adults, with 351 randomized to each of the three treatment arms.

11. ATLAS Trial (Antiretroviral Therapy as Long-Acting Suppression)

- **Objectives:** To establish whether switching to a long-acting injectable regimen of cabotegravir (CAB) + rilpivirine (RPV) administered every 4 weeks is non-inferior to continuing a standard daily oral ART regimen in virologically suppressed adults.
- **Findings:** The long-acting injectable regimen was found to be non-inferior, maintaining virologic suppression (HIV-1 RNA <50 copies/mL) at rates similar to daily oral therapy at Week 48.
- **Recommendations:** This trial was pivotal in demonstrating the efficacy and safety of a long-acting injectable regimen, leading to its approval as a new option for people with HIV who are virologically suppressed and prefer less frequent dosing.
- **Medications:** Cabotegravir (CAB), Rilpivirine (RPV) as long-acting intramuscular injections.
- **Publication:** *New England Journal of Medicine* (NEJM).
- **Year:** 2021 (primary 48-week results).
- **Participants:** Virologically suppressed, ART-experienced adults.

12. SApiT Trial (Starting Antiretroviral Therapy at Three Points in Tuberculosis)

- **Objectives:** To determine the optimal timing for initiating ART in patients co-infected with HIV and tuberculosis (TB) by comparing integrated treatment (starting ART during TB treatment) versus sequential treatment (starting ART after completing TB treatment).
- **Findings:** The trial's Safety Monitoring Committee halted the sequential treatment arm early due to a 56% lower mortality rate in the integrated treatment group, demonstrating a clear survival benefit to starting ART earlier during TB treatment.
- **Recommendations:** Provided critical evidence that integration of TB and HIV treatment improves survival, directly influencing global guidelines to recommend earlier ART initiation in co-infected patients.
- **Medications:** Standard TB treatment, plus various ART regimens including ddI, 3TC, and others.
- **Publication:** *South African Medical Journal*.
- **Year:** 2010.
- **Participants:** 645 patients co-infected with TB and HIV.

13. CAMELIA Trial (Cambodian Early vs. Late Introduction of Antiretrovirals)

- **Objectives:** To compare the impact on survival of initiating ART early (2 weeks after starting TB treatment) versus late (8 weeks after starting TB treatment) in severely immunosuppressed (CD4+ count ≤ 200 cells/mm³), treatment-naïve adults with newly diagnosed TB.
- **Findings:** Early ART initiation led to significantly improved survival. Mortality was 18% in the early arm vs. 27% in the late arm.
- **Recommendations:** The trial recommended that ART should be started as early as 2 weeks after initiating TB treatment in severely immunosuppressed HIV-TB co-infected patients to maximize survival benefit.
- **Medications:** A standard TB treatment regimen, plus an ART regimen of stavudine (d4T), lamivudine (3TC), and efavirenz (EFV).
- **Publication:** *BioMed Central (BMC) Proceedings*.
- **Year:** 2011.
- **Participants:** 661 HIV-infected adults with newly diagnosed TB and a CD4+ count of 200 cells/mm³ or less.

14. HVTN 702 Trial (Uhambo)

- **Objectives:** To evaluate the safety and efficacy of a prime-boost HIV vaccine regimen (ALVAC-HIV + Bivalent Subtype C gp120/MF59) for preventing HIV-1 infection in adults in South Africa.
- **Findings:** The trial was stopped early for futility in 2020 when an interim analysis showed the vaccine was safe but did not prevent HIV infection; there was no significant difference in infection rates between the vaccine and placebo groups.
- **Recommendations:** The study confirmed that despite promising earlier work, this specific vaccine approach was not effective, underscoring the immense scientific challenge of developing a preventative HIV vaccine.
- **Medications:** The intervention was a non-medication, investigational vaccine consisting of a canarypox vector-based vaccine (ALVAC-HIV) and a bivalent subtype C gp120 protein subunit vaccine with MF59 adjuvant.
- **Publication:** *New England Journal of Medicine (NEJM)* (expected); results widely reported in press releases.
- **Year:** 2020 (results announced).
- **Participants:** 5,400 HIV-uninfected, sexually active men and women aged 18 to 35 in South Africa.

15. TEMPRANO Trial (ANRS 12136)

- **Objectives:** To assess the benefits and risks of starting ART immediately versus deferring it based on prevailing WHO criteria, and to evaluate the efficacy of a 6-month course of isoniazid preventive therapy (IPT), in HIV-infected adults with CD4 counts $<800/\text{mm}^3$.
- **Findings:** Both immediate ART and 6-month IPT independently led to a lower risk of severe HIV-related illness or death compared to deferred ART and no IPT, even among participants with high CD4 counts (≥ 500 cells/ mm^3). Immediate ART reduced the risk of severe illness by 44%.
- **Recommendations:** The trial provided evidence from an African setting to support the global "treat-all" strategy and the use of IPT in settings with a high burden of tuberculosis.
- **Medications:** Immediate ART used various regimens; the preventive therapy was a 6-month course of isoniazid (INH).
- **Publication:** *New England Journal of Medicine* (NEJM).
- **Year:** 2015.
- **Participants:** HIV-infected adults with CD4 counts $<800/\text{mm}^3$.

16. VISCONTI Study (Viro-Immunological Sustained COntrol after Treatment Interruption)

- **Objectives:** To characterize the rare individuals known as "post-treatment controllers" (PTCs) who are able to maintain long-term virological remission after discontinuing ART, a state sometimes described as a "functional cure".
- **Findings:** The study identified and characterized the first 14 PTCs, showing they had all started ART very early during primary HIV infection and had been on treatment for a median of 3 years before stopping. A 2025 study further identified specific immunogenetic markers linked to natural killer (NK) cell function that are associated with this post-treatment control.
- **Recommendations:** The study demonstrated that long-term remission from HIV is possible and suggested that very early and sustained treatment may be a key factor, opening the door for research into strategies aimed at achieving a functional cure.
- **Medications:** Not a specific drug trial; participants had received various standard ART regimens during their treatment phase.
- **Publication:** *PLoS Pathogens* (2013); *Med* (2025 genetic study).
- **Year:** 2013 (initial findings); 2025 (genetic study).
- **Participants:** A cohort of 30 post-treatment controllers who had initiated ART early in infection.